

CRISPR System Homology and Evolution in the Paraburkholderia Symbionts of Dictyostelium discoideum

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Introduction:

Three species of bacteria from the Paraburkholderia genus *Paraburkholderia bonniea*, *Paraburkholderia hayleyella*, and *Paraburkholderia agricolaris*, have been found to engage in a facultative symbiotic relationship with the soil-dwelling amoeba *Dictyostelium discoideum* (Brock et. al. 2018). This symbiosis is valuable for understanding host-pathogen interaction as well as the evolutionary mechanisms and genomic effects of such a relationship (Bozzaro and Eichinger 2011, Noh et al 2022). The interaction of *D. discoideum* and infecting *Paraburkholderia* is mediated by many complex genetic and environmental factors, and research on this system has pointed to bacteriophages as an important driver of host-symbiont interaction (Clokier et. al. 2011). Recent work across diverse eukaryote–microbe symbiotic systems shows that phages can actively shape partnerships by modulating bacterial community composition, influencing host immunity, and generating rapid evolutionary change. These phage-mediated effects highlight that *D. discoideum*–*Paraburkholderia* associations likely depend not only on amoeba–bacteria dynamics, but also on the ecological and immunological conditions that determine phage access and persistence within the host (DiSalvo et. al. 2025, Xu et. al 2016). The symbiotic interactions of *D. discoideum* and these select *Paraburkholderia* species is in reality a tripartite system, in which bacteriophages also exhibit significant interaction with both the host and the infecting bacteria (DiSalvo et. al. 2025). Given that the phage community associated with *D. discoideum* infection differs from the phages that infect free-living *Paraburkholderia*, bacteria residing in the host may face distinct phage pressures, potentially driving alternative evolutionary outcomes. (DiSalvo et. al. 2025).

Bacteria possess an arsenal of innate and adaptive defense mechanisms intended to protect against bacteriophage infection and foreign genetic elements (Van Houte et. al 2016). Among these systems, CRISPR-Cas is notable as it has been found to function off of a genetic ‘memory’ of past phage interactions using phage-derived sequences, allowing for adaptive immunity to infection (Hillie and Charpentier 2016). Therefore, understanding the presence and functionality of CRISPR-Cas in *Paraburkholderia* symbionts is vital to fully understand the effects of bacteriophage pressure on host-symbiont interaction and the long-term genomic implications of phage defense in this context. Additionally, research on the genetic basis for symbiont progression to vertical and obligate symbiosis in *Arsenophonus* concluded that the deactivation of CRISPR defense is a key driver of genome expansion in early-stage symbiosis (Siozios et. al. 2024). Therefore, the state CRISPR systems in *Paraburkholderia* symbionts could offer valuable insight on the evolutionary history of this symbiotic system.

Preliminary analysis of cas gene conservation in *Paraburkholderia* found an absence of cas genes across the well-documented species of the genus, with the exception of *Paraburkholderia bonniea* and *Paraburkholderia hayleyella*. This observation informed two hypotheses for the presence of CRISPR associated genes in the two symbiont species: 1. CRISPR elements were acquired independently by

P.bonniea and *P.hayleyella* via horizontal gene transfer -. 2. CRISPR elements were acquired by a common ancestor of *P.bonniea* and *P.hayleyella* -.

Methods:

CRISPR Feature Identification:

All CRISPRCasFinder analysis was performed on complete genome assemblies. The set of Paraburkholderia taxa was composed of complete genomes from 23 strains of *P. hayleyella*, 16 strains of *P. bonniea*, and 9 strains of *P.agricolaris*. All genomes were directed by locating the chromosomal replication initiator dnaA sequence and initiator replication protein in each assembly's contig and reorienting each contig based on these genes using Circlator (Noh et. al. 2022).

All genomes were run on CRISPRCasFinder with default settings. Only CRISPR arrays assigned evidence level-4 were considered in this study. CRISPRCasFinder utilizes an algorithm to produce an entropy-based spacer conservation index which determines the degree of similarity between spacer sequences in a given array. True CRISPRs are expected to possess highly unique spacer sequences. Array assigned evidence level 3-4 are considered high likelihood candidates while evidence level 1-2 are considered potentially spurious matches (Couvin et. al. 2018). No evidence-level 3 arrays were located in any of the genomes, therefore only evidence level 4 arrays were chosen.

All CRISPR array features and genome locations were pulled from the .GFF output files produced by CRISPRCasFinder. Cas and Csy gene sequences and locations were taken from the rawCas .fasta output file. The visual diagram of *P. bonniea* and *P.hayleyella* CRISPR systems was created in Microsoft PowerPoint.

Direct Repeat and Cas/Csy Gene Alignment:

Consensus direct repeat (DR) sequences for all level-4 arrays located by CRISPRCasFinder were retrieved from the .GFF file associated with each genome. DR sequences from *P. bonniea* and *P.hayleyella* were aligned using MAFFT on default settings. The alignment visualization was generated by AlignmentVeiwer.org.

Cas1 and csy4 nucleotide sequences retrieved by CRISPRCasFinder were translated using the Bioinformatics.org translate tool. Both gene sequences were translated from the reverse strand in the first reading frame. Alignment of Cas1 and csy4 gene sequences from all *P. bonniea* and *P. hayleyella* strains was performed using MAFFT with 1000 iterations.

Cas Gene Phylogenies:

Cas1 and Csy4 sequences from *P. Hayleyella* strain number 11 were chosen as the representative query sequence. Cas1 and Csy homologs were found using the fast.genomics web tool with the selected option to return only "good" hits. The fast.genomics output was then filtered down to only non-duplicate hits for species within the Burkholderiaceae and Enterobacteriaceae families. Using the protein accession ID's retrieved by fast.genomics, a BASH script utilizing NCBI entrez-direct to retrieve the sequence associated with each ID and format them into a .fasta file. Cas1F and Csy4F homolog

sequences were aligned using MAFFT with 1000 iterations. Phylogenetic gene trees were generated using IQ-TREE on default settings. Tree visualization and aesthetic modification were performed using the R package 'ape'.

Spacer Analysis:

Spacer sequences belonging to all evidence-level 4 arrays from all *P. bonniea* and *P. hayleyella* strains were retrieved from the .GFF files associated with each strain produced by CRISPRCasFinder. This was achieved using a Python script which retrieved all spacer sequences from each evidence level-4 array, labeled them by strain and spacer number, and created a .fasta output. These files were concatenated together to form a .fasta file containing all spacers from Array A across all *bonniea* and *hayleyella* strains. An additional file containing only Array B spacers from all *bonniea* and *hayleyella* strains was created using the same Python script.

Blast databases were created from the .fasta spacer files using command-line makeblastdb with default settings. To detect shared spacer between strains, an all-against-all nucleotide BLAST was performed on the databases using the spacer .fasta files as queries. Only sequences with 98% or higher sequence identity were considered to be matches.

Spacer sharing between strains was assessed using Python by counting the number of spacer matches associated with each strain. Spacer sharing networks were visualized using the networkX Python package with the 'spring_layout' plotting method.

Additional taxa containing a homologous CRISPR array to *P. bonniea* and *P. hayleyella* were found by querying the CRISPRCasDB direct repeat database using the *P. hayleyella* consensus DR sequence for array A. Only strains found to have a consensus DR sequence identical to either *P. bonniea* or *P. hayleyella* and belonging to the Burkholderiaceae family were included. Complete genomes for these species were downloaded from the NCBI RefSeq database and were run on CRISPRCasFinder with the previously described method. For genomes with multiple array copies with the same DR, all spacers were included. These spacers were combined with the *bonniea* and *hayleyella* array A spacers and a new blast database was created using the previously described method. Only sequences with 90% or greater sequence identity were considered matches to account for higher likelihood of accumulated mutations in more-distantly related taxa. The spacer sharing network was visualized using the same methods.

Results:

Analysis of symbiont genomes with CRISPRCasFinder located a type I-F cas gene cluster and two evidence-level 4 CRISPR arrays in all 16 strains of *P. bonniea* and 23 strains of *P. hayleyella* (Couvin et. al. 2018). All CRISPR features were found on chromosome 1 in both genomes. Analysis of 9 *P. agricolaris* strain genomes with CRISPRCasFinder revealed no cas cluster or CRISPR arrays greater than evidence-level 1 on either chromosome. In *P. bonniea* and *P. hayleyella*, the identified *cas* genes are located on the reverse strand. The *cas* genes of both species were found to be in a contiguous cluster, located at 3038786bp-3047772bp in *P. bonniea* and 2044128bp-2053245bp in *P. hayleyella* (Figure 1). In

both species, a level-4 array was identified directly downstream of the cas cluster, separated from the cas6f CDS by a ~150bp leader sequence (Figure 1). This array, due to its direct proximity to the cas coding sequences, was labeled “Array A”. In all 16 strains of *P. bonniea* examined, a second level-4 array was identified 412kbp upstream of Array A. In all 23 strains of *P.hayleyella* examined, a second level 4 array was identified 1.16mbp downstream of Array A. These arrays were labeled “Array B” due to their extended distance from the cas cluster in both species' genomes (Figure 1).

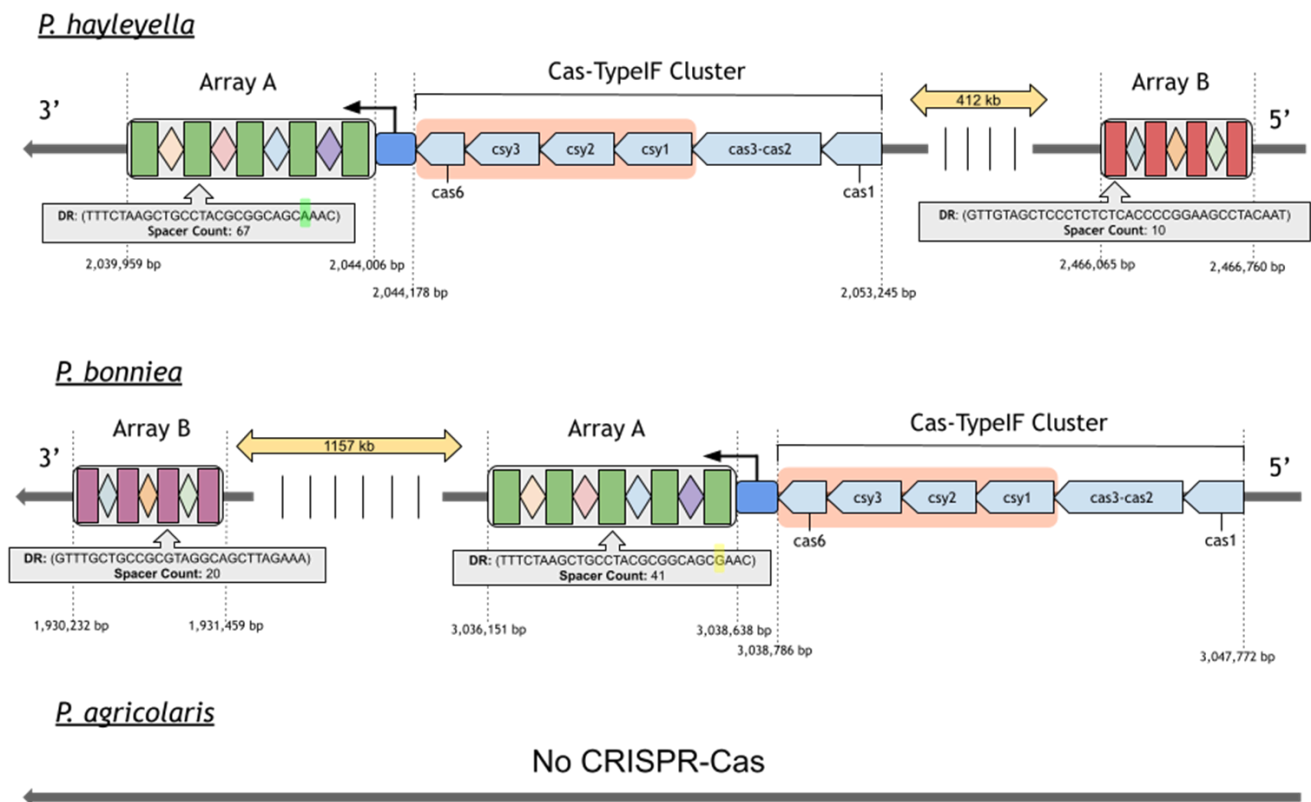


Figure 1. Graphic representation of CRISPRCas systems found in Paraburkholderia symbionts of *D. discoideum*. *P. hayleyella* strain no.11 and *P. bonniea* strain no.395 genomes were used as representative strains. All CRISPR array and Cas gene features were identified using CRISPRCasFinder with the default settings. Only CRISPR arrays identified as evidence level 4 are represented and considered to be “true” CRISPRs. Within the annotated cas cluster, the orange region highlights the cas genes associated with DNA interference, known as the effector complex.

The consensus direct repeat (DR) sequence for Array A in *P.bonniea* is 28bp in length and is identical across all sampled *P. bonniea* strains (Figure 1). The DR sequence for Array A in *P. Hayleyella* was also found to be 28bp long and identical across all sampled *P.hayleyella* strains (Figure 1). An alignment of the *bonniea* and *hayleyella* DR sequences revealed 96.4% sequence identity, with a difference of a

single nucleotide substitution. The *P.bonniea* DR sequence possesses a guanine at the 25th nucleotide position while the *P.hayleyella* DR sequence possesses an adenine at that position (Figure 2). The consensus DR sequence for Array B in *P. bonniea* is equal in length to the Array A DR. The reverse complement of the Array B DR sequence differs from the Array A DR by the same single A to G nucleotide substitution, making the Array B DR reverse complement identical to the Array A DR in *P. hayleyella* (Figure 1).

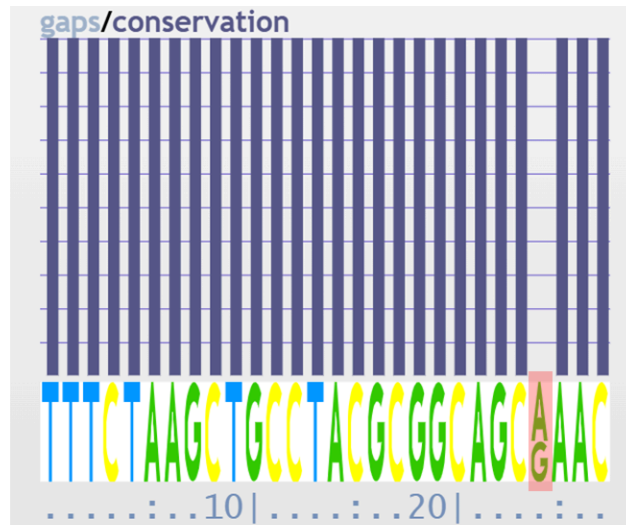


Figure 2. Consensus sequence from multiple sequence alignment of *P. bonniea* and *P. hayleyella* Array A direct repeat (DR). Consensus DR sequences from 16 *P. bonniea* and 23 *P. hayleyella* strains were aligned with MAFFT on default settings. Alignment and DR consensus visualized with AlignmentViewer software. Single nucleotide substitution highlighted in red.

The *cas1* gene sequences retrieved from bh11 and bb395 genomes bear 78.2% sequence identity while the *csy4* sequences bear 78.8% sequence identity. Both gene sequences were found to be identical among strains of the same species. Phylogenetic analysis of *cas* type I-F homologs of *P. bonniea* and *P.hayleyella*, from a range of close and distant relatives, shows that the *cas* gene sequences of *P. bonniea* and *P.hayleyella* are more similar to each other than to any of the other related taxa (Figure 3A). Additionally, the *cas1/csy4* sequences are more closely related to those of other Burkholderiaceae than to those of other families (Figure 3A-B). A phylogeny of aligned *cas1f* homologs from the Enterobacteraceae and Burkholderiaceae families show that *cas1f* protein sequences in *P.bonniea* and *P. hayleyella* are most similar to homologous sequences found in species from the Burkholderia and Caballeronia genera (Figure 3A). The same relationship is shown in a phylogeny of aligned *cys4f* homologs from Enterobacteraceae and Burkholderiaceae (Figure 3B). *Csy4* gene sequences for bh11 and bb395 genomes retrieved by CRISPRCasFinder show 78.2% sequence identity and 80% protein sequence identity.

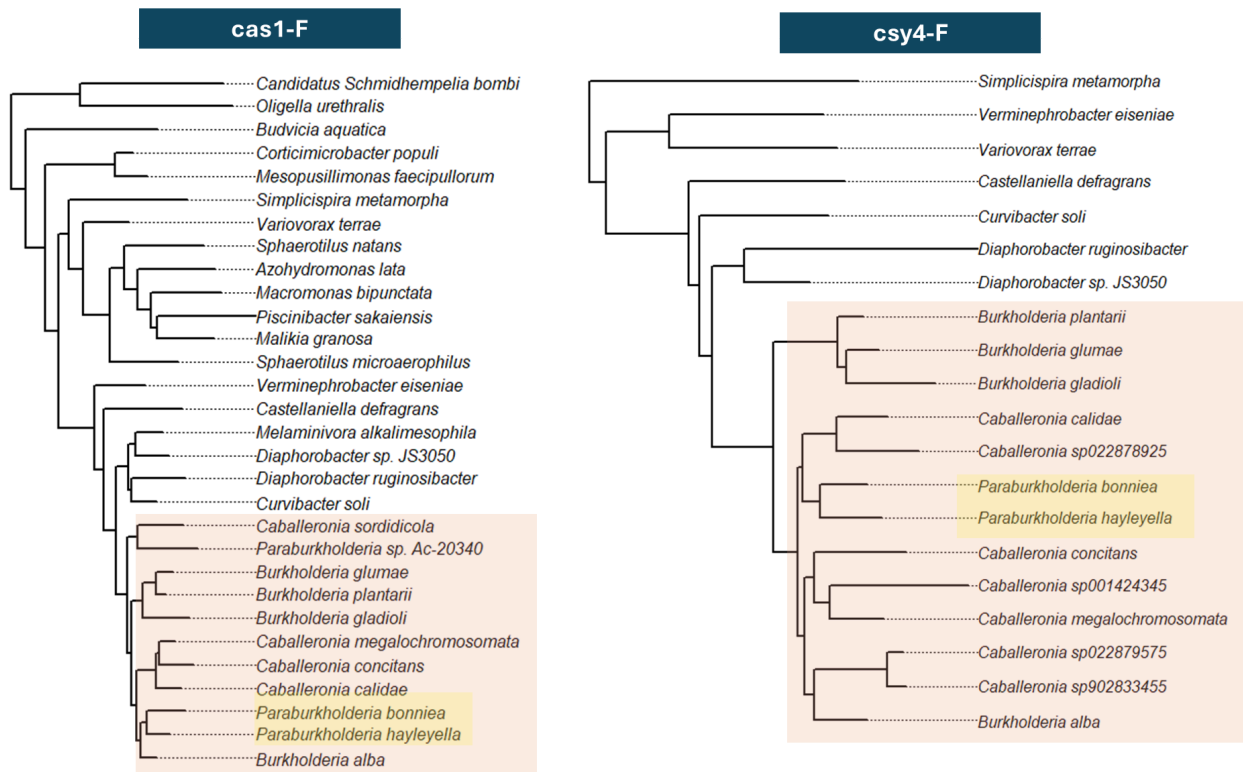


Figure 3. Phylogenetic trees of cas protein homologs in Burkholderiaceae and Enterobacteriaceae species. Homologs identified with fast.genomics software maintaining only “good” hits. Alignment of protein sequences performed using MAFFT with 100 iterations. Trees generated with IQ-TREE on base parameters A) Phylogenetic tree of aligned protein homologs of CRISPR Type I-F adaptation module gene *cas1*. B) Phylogenetic tree of aligned protein homologs of CRISPR Type I-F effector module gene *csy4*.

Analysis of Array A in *P. bonniea* revealed that 14 of the 16 sampled strains possess an equal number of 41 spacer sequences. The two remaining *bonniea* strains, bb859 and bb863, were found to possess 33 spacers in Array A. In *P. hayleyella*, Array A was found to contain an equal 67 spacers across all 23 sampled strains. Array B in *P. bonniea* contains 20 spacers, with the exception of bb859 and bb863, which contain 16. In *P. hayleyella*, Array B was found to contain an equal 10 spacers across all 23 strains.

An all-against-all nucleotide BLAST of Array A spacers from all sampled *P. bonniea* and *P. hayleyella* strains, with a minimum threshold of 98% sequence similarity, found a high degree of spacer conservation between strains of the same species but no spacers shared between strains belonging to different species (Figure 4). *P. bonniea* strains share 40 of 41 spacers in Array A with the exception of strains bb859 and bb863. These strains share all 33 of their spacers with each other, but share only 5 spacers with the other 14 *bonniea* strains. *P. hayleyella* strains share all 67 spacers in Array A (Figure 4).

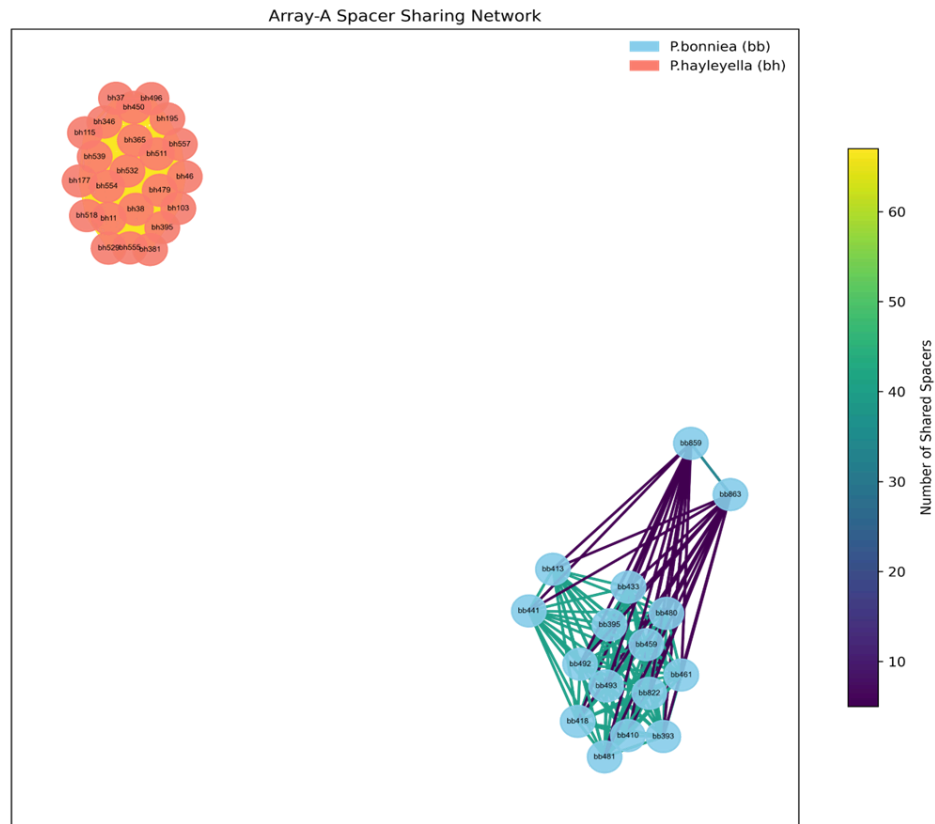


Figure 4. Force-directed network representing Array A spacer sequences shared between *P. hayleyella* and *P. bonniea* strains. Spacer sequence matches were determined using an all-against-all blastn with a percent identity threshold of 98%. The color of the connecting edges corresponds to the number of spacers shared between strain arrays. Node color has no relationship to shared spacer count. Spacer sequences were taken from the contents of the Array A CRISPR cassette in 23 *P. hayleyella* and 16 *P. bonniea* strains.

An all-against-all nucleotide BLAST of Array B spacers from all sampled *P. bonniea* and *P. hayleyella* strains, with a minimum threshold of 98% sequence similarity, also showed a high degree of spacer conservation between strains of the same species but no spacers shared between strains belonging to different species (Figure 5). *P. bonniea* strains share all 20 Array B spacers with the exception of bb859 and bb863. These strains share all 16 of their spacers with one another, but only 13 of these are shared with other *bonniea* strains. All strains except bb859 and bb863 were found to contain two identical spacers sequences within the array, indicating a duplicated spacer. *P. hayleyella* strains share all 10 spacers in Array B.

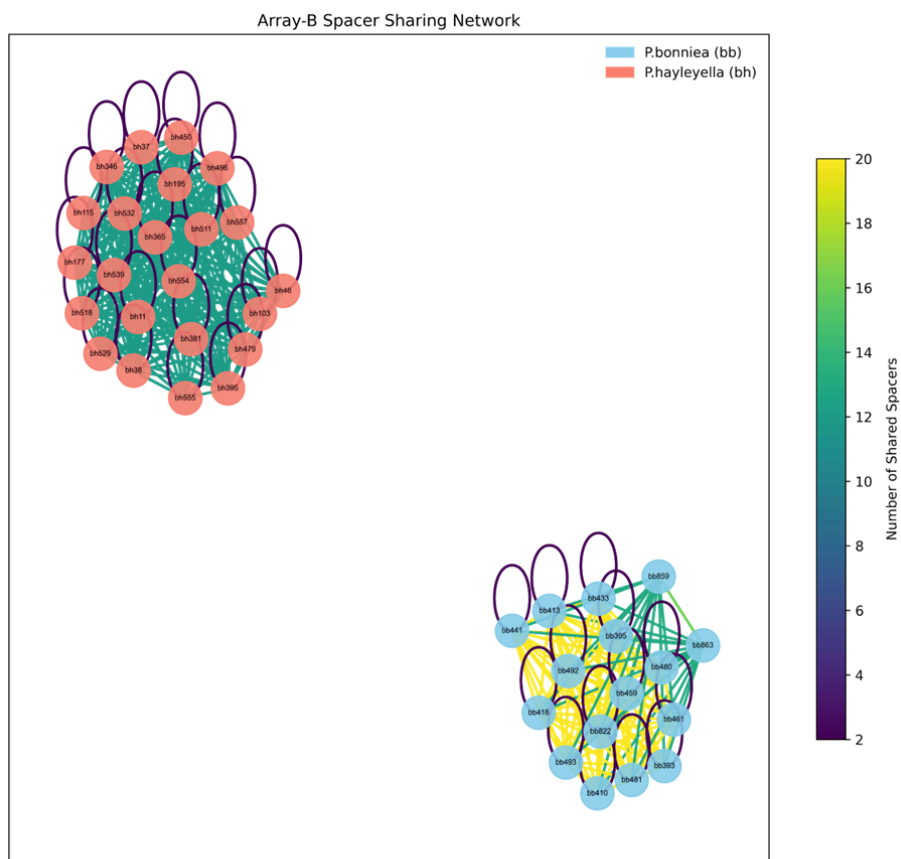


Figure 5. Force-directed network representing Array B spacer sequences shared between *P. hayleyella* and *P. bonniea* strains. Spacer sequence matches were determined using an all-against-all blastn with a percent identity threshold of 98%. The color of the connecting edges corresponds to the number of spacers shared between strain arrays. Looped edges represent repeated spacers within a CRISPR array. Node color has no relationship to shared spacer count. Spacer sequences were taken from the contents of the Array A CRISPR cassette in 23 *P. hayleyella* and 16 *P. bonniea* strains.

A nucleotide BLAST search on the CRISPRCasFinder DR database, using *P. bonniea* and *P. hayleyella* Array A DR sequence as a query, found several species from the Burkholderiaceae family possessing evidence-level 4 CRISPR arrays with highly similar DR sequences. Taxa previously shown to possess high *cas* gene homology to *P. bonniea* and *P. hayleyella* were selected for spacer comparison. These species were *Burkholderia plantarii*, *Burkholderia gladioli*, *Caballeronia concitians*, and *Caballeronia cladiae*. Each strain included from these species possesses at least 1 evidence-level 4 CRISPR array with a DR sequence >90% identical to the *bonniea* and *hayleyella* Array A DR. An all-against-all nucleotide BLAST of *P. bonniea* and *P. hayleyella* spacers with all additional species spacers, with a similarity threshold of 90%, found no spacer matches between species (Figure 6). Three *B. plantarii* strains (ATTC, KACC, PG1) were found to share a single spacer from arrays containing 33, 29, and 39 spacers respectively (Figure 6).

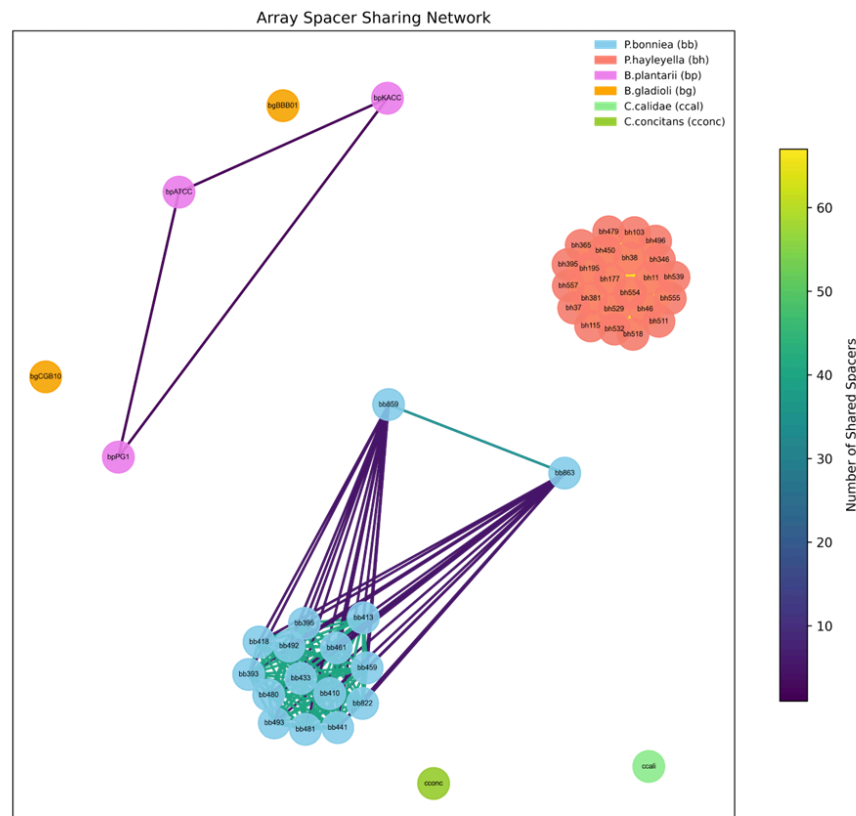


Figure 6. Force-directed network representing Array A spacer sequences shared between *P. hayleyella*, *P. bonniea* and related species from the Burkholderiaceae family. Spacer sequence matches were determined using an all-against-all blastn with a percent identity threshold of 98%. The color of the connecting edges corresponds to the number of spacers shared between strain arrays. Node color has no relationship to shared spacer count. Spacer sequences were taken from the contents of the Array A CRISPR cassette in 23 *P. hayleyella*, 16 *P. bonniea* strains, 3 *B. plantarii*, 2 *B. gladioli*, 1 *C. concitans*, and 1 *C. cladiae* strains.

Discussion:

The goal of this study was to investigate the presence of CRISPR/Cas defense systems among the three *D. discoideum* symbionts, assess their completeness, and develop an understanding of their evolutionary history within these genomes. The results of this study suggest that complete CRISPR/cas systems, including a continuous cas gene CDS cluster and an accompanying array, can be found in all strains of *P. bonniea* and *P. hayleyella*. No evidence was found to suggest the presence of a complete CRISPR/Cas system or cas pseudogenes in *P. agricolaris*. Within *P. bonniea* the CRISPR array directly associated with the cas cluster possesses a highly similar DR sequence to the corresponding array in *P. hayleyella*. Cas1 and csy4 protein sequences from *P. bonniea* and *P. hayleyella* were found to be more similar to one another than to other species within or outside of the Burkholderiaceae family. All-against-all comparison of the CRISPR array spacer content of *P. bonniea*, *P. hayleyella*, and closely

related taxa possessing a homologous array found no examples of spacers shared between species. However, among strains of *P.bonniea* and *P.hayleyella*, spacer content is highly conserved.

The findings of this study support the hypothesis that the CRISPR/Cas system found in *P. bonniea* and *P.hayleyella* was acquired from a common ancestor. DR sequences are typically well conserved in the arrays of functional CRISPRs, suggesting that the CRISPR/Cas systems found in *P. bonniea* and *P.hayleyella* are homologous. This hypothesis is further supported by the similarity of both adaptation and effector module genes within the cas cluster between both species. Prior research has highlighted that cas1 exhibits the highest high rates of conservation in homologous CRISPR systems and cys4 is essential to functional interference in Type I-F CRISPR systems (Cady et. al 2011, Xue et. al. 2019). The similarity of the cas1 protein encoded by *bonniea* and *hayleyella* further supports that their respective CRISPR systems share a common origin. Despite CRISPR cas systems frequently being transferred between bacterial genomes, the findings of this study do not support the hypothesis that the CRISPR type I-F system found in *P. bonniea* and *P. hayleyella* were independently acquired via recent HGT events. Rather, the findings support that CRISPR was acquired by a common ancestor of *P. bonniea* and *P. hayleyella* and has been maintained in each species.

First, the location of the system on chromosome two in both species suggests long-term integration into the genome, as recently transferred CRISPRs are often encoded on plasmids (Mohanraju et al 2022). Second, a protein BLAST search using cas1 and csy4 from *P. bonniea* and *P. hayleyella* across closely and distantly related taxa found no sequences more similar than those belonging to species in the Burkholderiaceae family. Gene trees generated using the cas1 and csy4 gene sequences present a highly similar structure as species trees for Burkholderiaceae (Moriuchi et. al. 2019). This suggests that these sequences did not originate in a distantly related bacterial genome, which would signal an HGT event. While this does not preclude the possibility of CRISPR being horizontally transferred from a species within Burkholderiaceae, the results of the spacer sequence analysis indicate that this is unlikely. The spacer content of an array corresponding to a functional CRISPR system will change over time as new spacers are integrated into the array and others are lost through mutation events (Kupczok & Bullback 2013, Gophna et al. 2015). Therefore, we would expect an array recently that has been acquired to share the majority of its spacers with the donor array. All-against-all comparison of spacers from *P.bonniea* and *P.Hayleyella* with Burkholderiaceae species found to possess a homologous array, revealed no shared spacers. This finding weakens the possibility that recent HGT events independently introduced CRISPR systems into *P. bonniea* and *P. hayleyella*, and instead supports the hypothesis that purifying selection acting on a CRISPR system inherited from a common ancestor is responsible for the homologous CRISPR and cas elements found in both species.

A number of factors also support the conclusion that the CRISPR systems found in *P. bonniea* and *P. hayleyella* are currently, or have recently been, functional. The protein sequences of cas1, and csy4 are identical among the 16 *P. bonniea* strains and the 23 *P. hayleyella* strains analyzed in this study. This unanimous conservation of these essential CRISPR genes suggests ongoing purifying selection among strains of these symbionts. This supports the notion that active CRISPR defense against bacteriophages, or possibly an alternative CRISPR function, confers a fitness benefit to *P.bonniea* and *P. hayleyella* that outweighs the high energy cost of maintaining CRISPR defense (Van Houte et. al 2016). If CRISPR was no longer functional, and therefore not providing a fitness advantage, we would expect to see some

variation in *cas1* or *csy4* between due to accumulated mutations. The absence of shared spacers between *P. bonniea* and *P. hayleyella* indicates that spacer acquisition has continued independently in each lineage, leading to divergent CRISPR arrays. At the same time, the observation that most strains within each species share identical spacer sets suggests that this diversification occurs on relatively long evolutionary timescales, which has been observed in other CRISPR-containing bacteria (Gophna et al. 2015). A lack of spacer sharing between non-Paraburkholderia species with homologous CRISPR arrays suggests that the two symbiont species experienced differential exposure to bacteriophages. Additionally, among the three available *B. plantarii* strains, a much lower degree of spacer sharing was observed. This observation is weakened by the lack of additional strains containing homologous arrays, but may support the assertion that interaction with *D. discoideum* uniquely affects spacer conservation in *P. bonniea* and *P. hayleyella*.

Living within *D. discoideum* may expose *P. bonniea* and *P. hayleyella* to a narrower and more stable phage community than free-living bacteria encounter. This is supported by prior study of spacer content in CRISPR containing symbiotic bacteria of the *Arsenophenous* genus which found that a lack of spacer sharing between symbiotic clades indicated differences in the mobile genetic element environment the bacteria are exposed to (Siozios et. al. 2024). A restricted set of viral pressures due to a symbiotic lifestyle within *D. discoideum* would limit opportunities to acquire new spacers, helping maintain the high spacer conservation observed between strains in *P. bonniea* and *P. hayleyella*. In contrast, *B. plantarii* likely experiences a more diverse and dynamic phage pool, leading to greater spacer turnover and lower levels of shared CRISPR content. These conditions may also help to explain why *P. bonniea* and *P. hayleyella* possess such different spacer content when comparing between species, despite their close relatedness. Host-imposed limits on phage access, together with species-specific differences in phage susceptibility, may create highly distinct viral environments for each symbiont. However, the precise factors underlying the minimal spacer sharing between *P. bonniea* and *P. hayleyella* remain unresolved and warrant further investigation.

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